

2026

COMPUTER SCIENCE — MINOR

Paper : MN-3

(Data Structure)

Full Marks : 75

*The figures in the margin indicate full marks.**Candidates are required to give their answers in their own words as far as practicable.*1. Answer **any five** questions :

2×5

- (a) Define an Abstract Data Type (ADT). Give an example.
- ~~(b)~~ Explain sparse matrix. Mention one method of representing a sparse matrix.
- ~~(c)~~ Differentiate between arrays and linked lists (any two points).
- (d) State advantage of circular linked list over singly linked list.
- ~~(e)~~ Define stack. Mention any two applications of stacks.
- (f) Explain tail recursion.
- ~~(g)~~ Differentiate between linear search and binary search based on time complexity.
- ~~(h)~~ Define hashing. What is meant by collision in hashing?

Section-A

Answer **any three** questions.

- ~~2.~~ Write an algorithm for insertion and deletion operations in a singly linked list. Explain with a suitable example. 3+2
- ~~3.~~ Convert the infix expression $(A+B)*(C-D)/E$ into postfix form and write the algorithm for infix to postfix conversion using stack. 5
- 4. Explain sparse matrix representation and polynomial representation using arrays. 5
- ~~5.~~ Explain array representation of stack with PUSH and POP operations. 5
- 6. Explain Binary Search Tree (BST). Describe searching and insertion operations in BST with suitable examples. 1+4

Please Turn Over

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Section - B

Answer *any five* questions.

7. (a) Construct the following queue of characters where queue is a circular array which is allocated six memory cells.
 FRONT=2 REAR=4 QUEUE : _A, C, D, _
 Describe the Queue as the following operations take place :
 (i) F is added to the Queue
 (ii) Two letters are from the Queue
 (iii) K,L,M are added to the Queue
 (iv) Two letters are deleted from the Queue
 (v) R is added to the Queue
 (vi) One letter is deleted from the Queue. 5+5
- (b) Write an algorithm for inorder traversal of a threaded binary tree. 5+5
8. (a) Write a recursive algorithm (pseudocode) for sorting an array using straight merge sort technique. 6+4
 (b) Discuss the drawback of linear probing and its possible remedy. 6+4
9. (a) Define heap. Construct a max-heap with the following array of elements :
 17, 25, 10, 7, 16, 19, 23, 42, 37, 4, 33, 1, 5, 11, 21
 Show with explanation, the different intermediate stages in the process of forming the heap.
 (b) Show the result of inserting 5, 3, 6, 8, 11, 4, 7, 9 into an initially empty binary search tree. Then show the result of deleting the root. Insertions and deletion should be shown in step by step manner. (1+4)+5
10. (a) Given input {4371, 1323, 6173, 4199, 4344, 9679, 1989} and a hash function $h(X) = X \bmod 10$, show the result of the following :
 (i) Open addressing hash table using linear probing.
 (ii) Open addressing hash table using quadratic probing.
 (b) Explain the terms row-major order and column-major order with an example. 6+4
11. (a) "Recursion is not always advantageous to use instead of iteration."— Justify with a suitable example.
 (b) Write an algorithm (pseudocode) for insert and delete operations of a circular queue represented by an array. 4+6
12. (a) What do you mean by priority queue? Explain how the insert and delete operations are performed in a priority queue.
 (b) Write an algorithm (pseudocode) to add two single variable polynomial using linked list. 5+5

(3)

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43. (a) Show all the steps to sort the following data using quicksort (ascending order) :

25,15, 30,10,40,19, 21, 34, 10, 29, 4

5+5

- (b) Write an algorithm (pseudocode) to reverse a singly linked list.
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